

BANCROFT, ON, Canada

WMO: 712940

Lat: 45.0716N

Lon: 77.8799W

Elev: 1085

StdP: 14.13

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.7	-13.4	-25.6	1.4	-18.9	-19.2	2.0	-12.2	14.1	18.0	12.9	16.5	1.0	0	0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.8	85.1	69.3	82.2	67.7	79.5	66.2	72.5	80.2	70.7	77.9	69.0	75.7	6.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.2	115.9	75.7	68.5	108.8	73.7	66.8	102.7	72.2	36.9	79.6	35.3	78.2	33.9	75.6	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	11.4	9.9	-28.9	89.5	5.1	2.8	-32.6	91.5	-35.5	93.1	-38.4	94.7	-42.1	96.7		
(5)			-28.6	75.4	5.5	2.1	-32.5	76.9	-35.8	78.2	-38.9	79.4	-42.9	80.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.0	13.5	16.3	26.0	39.2	52.4	61.0	65.2	63.3	56.1	44.3	32.1	20.9		
	DBStd	20.41	13.78	11.91	11.60	8.40	8.10	6.79	5.69	5.75	7.83	8.39	9.57	11.71		
	HDD50	4921	1133	942	747	339	67	4	0	0	29	220	537	903		
	HDD65	8952	1598	1362	1211	775	400	156	68	101	285	642	986	1368		
	CDD50	1634	0	0	2	15	140	335	472	414	211	44	1	0		
	CDD65	188	0	0	0	0	8	38	75	50	16	1	0	0		
	CDH74	2353	0	0	1	15	188	483	893	563	199	11	0	0		
	CDH80	515	0	0	0	2	39	110	211	114	39	0	0	0		
	WSAvg	4.0	4.2	4.2	4.7	4.8	4.2	3.7	3.5	3.1	3.3	4.0	4.1	4.0		
	PrecAvg	35.3	2.4	2.2	2.4	2.5	3.1	3.7	3.0	3.2	3.7	3.0	3.0	2.9		
(15) Precipitation	PrecMax	45.1	4.5	3.9	6.2	4.8	5.7	7.9	5.4	7.2	9.8	5.5	5.2	5.6		
	PrecMin	26.0	1.0	0.7	0.7	0.4	1.1	0.7	0.9	1.5	1.3	0.3	1.3	0.6		
	PrecStd	5.0	0.9	1.0	1.4	1.3	1.3	2.0	1.2	1.2	2.0	1.3	0.9	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.1	47.7	63.9	75.9	84.8	86.8	89.0	88.0	85.0	75.2	60.5	51.4		
		MCWB	46.2	41.1	52.2	58.8	65.8	70.3	72.0	70.7	69.4	63.2	51.9	49.5		
	2%	DB	40.6	42.0	53.3	67.6	79.0	83.3	85.2	82.9	79.5	69.0	55.1	44.4		
		MCWB	39.2	38.4	45.6	53.2	62.2	68.7	70.0	68.4	67.3	59.2	51.5	42.7		
	5%	DB	36.0	36.9	47.5	61.3	74.6	79.8	81.9	79.8	75.3	63.5	51.0	39.6		
		MCWB	34.6	34.3	40.7	48.9	59.2	66.0	67.6	66.8	64.7	57.2	47.2	38.0		
	10%	DB	32.8	33.8	42.2	55.9	70.0	76.0	79.2	76.9	71.1	59.1	46.8	35.5		
		MCWB	31.5	31.7	37.0	45.7	57.2	63.8	66.5	65.6	63.1	54.4	43.7	34.0		
	0.4%	WB	46.7	44.2	55.9	61.7	69.1	73.7	76.4	73.8	72.3	66.1	55.8	50.4		
		MCDB	47.0	45.2	62.2	71.0	79.0	82.4	85.2	81.6	79.5	69.6	57.6	51.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.4	38.7	47.1	56.4	66.0	71.1	72.9	71.5	69.7	61.7	52.0	42.8		
		MCDB	40.5	41.3	51.9	64.1	75.3	79.6	80.3	78.6	76.2	66.4	54.4	44.1		
	5%	WB	34.8	34.7	41.7	50.8	63.0	68.9	70.7	69.8	67.4	58.3	47.7	38.1		
		MCDB	35.8	36.7	46.3	57.3	70.3	76.0	77.7	76.0	72.6	62.9	50.2	39.4		
	10%	WB	31.7	31.9	37.5	46.9	59.5	66.5	68.7	67.9	64.7	54.6	43.9	34.3		
		MCDB	32.6	33.6	41.7	54.8	66.6	72.4	75.8	73.2	69.7	58.2	46.4	35.3		
	5% DB	MDBR	20.8	22.7	22.2	22.9	25.7	23.4	23.8	23.8	24.2	19.4	16.6	16.9		
		MCDBR	17.7	22.3	27.7	34.9	33.9	28.6	27.5	27.4	28.3	27.3	23.2	16.2		
		MCWBR	16.9	18.9	18.8	21.0	17.7	14.9	14.0	14.4	16.4	18.1	18.5	15.0		
		MCWBR	17.3	20.3	24.8	28.0	26.1	23.4	22.5	21.7	23.5	24.2	20.6	15.2		
(40) Clear-Sky Solar Irradiance	taub		0.274	0.285	0.305	0.340	0.356	0.396	0.396	0.385	0.358	0.339	0.300	0.270		
		taud	2.375	2.344	2.439	2.431	2.426	2.321	2.331	2.359	2.433	2.483	2.533	2.440		
	Ebn at Noon		268	287	296	292	290	277	275	274	273	264	259	259		
		Edh at Noon	23	29	31	34	36	40	39	36	31	26	21	20		
(44) All-Sky Solar Radiation	RadAvg		429	720	1147	1425	1670	1813	1835	1598	1263	744	450	327		
	RadStd		51	77	104	137	161	147	143	87	114	71	53	39		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (36 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page